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Information message

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Champions: Submission #475

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Project category	Basic information	**Project description**	Complete

Information message

PLEASE NOTE: This part of your application will be the basis of the experts' and jury's assessment of your project. Make sure to use the following headings as well as ANNEX 8 in the **Guide to Applicants** to help you throughout the application process..

Close

We look forward to reading your application! When answering the questions, please provide a clear and comprehensive

description of your project that highlights its impact, its relevance to the category you are applying for, and how it reflects the values and working principles of the New European Bauhaus.

Detailed description of the project

TELL US ABOUT YOUR PROJECT *

Please provide a comprehensive summary of your project. Make sure to focus on already achieved results as well as long term impact(s).

Cryptographic Beings is a bio-hybrid machine that stores digital data in living algae, proposing an alternative to silicon-based computation where data permanence depends on ecological stewardship. The project addresses the growing environmental and philosophical costs of digital infrastructure by

Maximum 475 characters

TELL US THE STORY YOUR PROJECT *

How did your project come about and how was it implemented on the ground.

The project emerged from a research question: what would computation look like if it depended on living organisms rather than silicon? The observation that Marimo algae spheres naturally float when photosynthesising and sink in darkness suggested a biological mechanism for binary switching. This natural

Maximum 328 characters

Please indicate the main themes of your project with 5 key words. *

By “themes,” we mean the central topics, approaches, or intended impacts of your project (e.g. what it focuses on, how it works, or the change it aims to create). You may include subject areas, methods, target groups, outcomes, or values.

Where relevant, you may align your keywords with NEB priorities and values. Examples: “circular economy,” “community co-design,” “energy efficiency,” “social inclusion,” “nature-based solutions.”

- Bio-hybrid computing
- Ecological stewardship
- Art-science transdisciplinarity
- Living data infrastructure
- Nature-based solutions

ADHERENCE TO THE NEW EUROPEAN BAUHAUS

Exemplary character in terms of the three core values of the New European Bauhaus.

*For more information of these values, please refer to the **Guide to Applicants** and to the **Guide to Applicants for the International Prizes**.*

NEB value: Sustainability *

How does your project concretely address environmental, social, cultural, or economic challenges and opportunities? In what way can your project be considered exemplary in terms of sustainability?

Cryptographic Beings directly addresses the environmental cost of computation and data storage. By substituting silicon-based transistors with living photosynthetic organisms, it proposes an alternative substrate where data permanence depends on ecological stewardship rather than energy-intensive

Maximum 922 characters

NEB value: Aesthetics *

How does your project concretely contribute to making life more beautiful for people, communities and neighbourhoods – both in the everyday and in the longer term? How does your project contribute to creating a sense of belonging and care for the place? In what way can your project be considered exemplary in terms of aesthetics?

Thirty transparent tubes with luminous green algae spheres create a mesmerising kinetic display, with spheres slowly rising and sinking in choreographed patterns. The pace of photosynthesis, from minutes to hours per state change, creates a contemplative, meditative viewing experience that stands in stark

Maximum 1148 characters

NEB value: Inclusion *

How does your project concretely address challenges and opportunities related to accessibility and affordability for all? How does it promote inclusive governance systems, design for all principles, or new societal models? In what way can your project be considered exemplary in terms of inclusion?

The installation is built around universally observable phenomena: floating and sinking balls, light and darkness. This makes both the binary logic and the underlying biological processes immediately legible, regardless of technical background. The living organisms engage viewers across generations and

Maximum 861 characters

Exemplary character in terms of the three working

principles of the New European Bauhaus.

*For more information of these working principles, please refer to the **Guide to Applicants** and to the **Guide to Applicants for the International Prizes**.*

NEB working principles: Participatory *

How has your project involved citizens, communities and civil society organisations who benefit from your project, or are affected by it? What was their role and level of involvement? What has been the impact of their involvement in the project?

Each exhibition is configured as a live experiment where audience presence directly affects the installation. The warmth of visitors' bodies and the CO2 from their breathing subtly alter the algae's metabolic rate, shifting photosynthetic activity and influencing how quickly the spheres respond to light changes. The audience is

Maximum 868 characters

NEB working principles: Multilevel governance *

How were stakeholders at different levels (local, regional, national, and European) involved in the design and implementation of the project? What was their role and level of involvement? What was the added value of their engagement?

The work has been exhibited in major international art institutions and festivals, engaging general audiences with questions about technology and ecology. It has been published in specialised journals including IEEE, and presented at academic conferences, contributing to ongoing research on bio-hybrid systems. Partnerships

Maximum 852 characters

NEB working principles: Transdisciplinary *

What disciplines and/or knowledge fields (e.g architecture, design, engineering, social sciences, arts, education, digital technologies, environmental sciences) were reflected in the design and implementation of your project?

Cryptographic Beings sits at the intersection of several fields that rarely converge. The biological layer draws on algae ecology, photosynthesis research, and buoyancy physics. The computational layer engages with binary logic, data storage theory, and embedded systems engineering, while AI integration adds a dimension of

Maximum 955 characters

EU COMPETITIVENESS

BUSINESS MODEL *

Please describe the business model behind your project. What makes your project financially sustainable over time (e.g. activities generating revenues, public funding, donations, crowdfunding, specific financial instrument, etc.)?

Maximum 2000 characters

High potential for transferability and replicability to other geographies and/or sectors

SCALABILITY *

Which elements of your project could be replicated or transferred to other places, groups of beneficiaries, or contexts (e.g. aspects of the methodology, technology, processes, products, tools, or key learnings etc.)?

Maximum 2000 characters

Mandatory Documents

Please provide documents showing the results, outcomes and impacts of your project (e.g.: evaluation report, final project report, certification, publications, etc.) *

- ○ PDF
- ◉ Web link
- ○ Both

Web link *

Web link *

Add

This must be an external URL such as http://example.com

Please attach at least 6 high resolution images (or other types of visual representations, if images are not possible) to your application. *

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Allowed types: gif, jpg, jpeg, png.

If you would like to upload additional documentation, please upload or write below

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Declaration

- [ ] By submitting your application, you confirm that you are the copyright owner of the images provided and that they can be used by the European Commission for the purposes of promoting the New European Bauhaus on all platforms (websites, print, social media and other). *
- [ ] By ticking this box, you declare that you are not in in one or more of the exclusion situations foreseen under Article 136 of the Financial Regulation. *
- [ ] By submitting this application, I declare that I am not related to the entities subject to EU restricted measures under Article 29 of the Treaty on the European Union (TEU) and subject to Article 215 of the Treaty on the Functioning of the EU (TFEU)1, and entities covered by Commission Guidelines No 2013/C 205/05), which are not eligible to participate in any capacity. *
- [ ] By ticking this box, you declare that all the information provided in this form is factually correct, that you assume sole liability in the event of a claim relating to the activities carried out in the framework of the contest, that the proposed project has not been proposed for the New European Bauhaus Prizes 2026 in any other category or strand and that it has not been subject to any type of *

investigation, which could lead to a financial correction because of irregularities or fraud.

☐ By submitting your application, you understand that all the applications that meet the eligibility requirements will be shared for the purposes of the selection processes, and notably published on the secured platform https://prizes.new-european-bauhaus.europa.eu/ and for the purposes of the promotion of these on the New European Bauhaus website and/or other European Commission communication channels. In this sense, the applications would be widely available. Applicants should ensure that they present their projects, in such a way that they could be shared without giving rise to intellectual property related concerns. If your submission is selected as one of the finalists, it will additionally be shared for the purpose of the public vote that will take place. The European Union is granted a licence to use and share your application with the general public and the official external experts for the purposes of the selection process, including the voting. The European Union has the right to use the images and visual materials and the description provided in the application for communication purposes related to the contest and beyond. Rights granted comprise the right to store, reproduce, display, publish and communicate or distribute copies in electronic or digital format, including, but not only, through the internet. Unless you have disclosed your name, the European Commission has no obligation to share your name when using or disseminating your contribution to the public. The European Union cannot be held responsible in case any submitted project is found to infringe third party's rights. The European Union shall be neither responsible for the use that third parties may do of the applications or related content. *

☐ By submitting your application, you acknowledge that you read and understand that the Guide to Applicants of the NEB Prizes 2026, available online in the website of the NEB Prizes 2026, is an integral part of this application form and that it must therefore be observed when completing and submitting your application. *

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